The Relationship Between Poverty and Cardiovascular Heart Disease Mortality in Women Ages 35 and Above Across California Counties

Data and Methodology

1. **Research Question and Background**
   This study analyzes how poverty in women 35+ are associated with female heart disease mortality across California's 58 counties. Geographic analysis of cardiovascular heart disease can help identify counties with socioeconomic disadvantages. This study answers the question: To what extent are poverty rates in women ages 35+ associated with cardiovascular heart disease mortality rates across California counties?
2. **Data Sources and Study Area**
   The socioeconomic data comes from the U.S. Census Bureau 2020-2024 ACS (American Community Survey) 5 year table B17001 that looks at poverty status by sex and age. The cardiovascular health data comes from the CDC Heart Disease Mortality Data Among US Adults (35+) by state, territory, and county from 2022-2024 covering females, race, and county-level records with both datasets including all 58 counties.
3. **Variables**

Variable	Source	Population	Time period
Poverty rate	ACS B17001	Women (35+); percentage below the poverty threshold	2020-2024
Cardiovascular Heart Disease Mortality	CDC Heart Disease Mortality	Women (35+)	2022-2024

4. **Methods**

5. **Methodological LImitations**
   A positive county-level correlation doesn’t mean that women living in poverty have an increased chance of dying of cardiovascular heart disease or that poverty can cause mortality. ACS estimates have a sampling error because of county size, and county mortality rates also have a statistical uncertainty.

Results